

Module - 01

01. With usual notation, prove that $\tan \phi = r \frac{d\theta}{dr}$
02. With usual notations, prove that the pedal equation of the form $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$
03. Find the angle of intersection of the following curves:
 - (i) $r^n = a^n \cos n\theta$ & $r^n = b^n \sin n\theta$
 - (ii) $r^2 \sin 2\theta = 4$ & $r^2 = 16 \sin 2\theta$
 - (iii) $r = a(1 + \cos \theta)$ & $r^2 = a^2 \cos 2\theta$
 - (iv) $r = a(1 + \sin \theta)$ & $r = a(1 - \cos \theta)$
 - (v) $r = 2 \sin \theta$ & $r = 2 \cos \theta$
 - (vi) $r = 4 \sec^2 \left(\frac{\theta}{2} \right)$ & $r = 9 \operatorname{cosec}^2 \left(\frac{\theta}{2} \right)$
04. Find the pedal equation of the following curves:
 - (i) $r = \frac{2a}{1 + \cos \theta}$
 - (ii) $r(1 - \cos \theta) = 2a$
 - (iii) $r^2 = a^2 \cos 2\theta$
05. Derive Radius of Curvature in the cartesian form.
06. Find the radius of curvature of the parabola $y^2 = 4ax$ at the point (a, a) .
07. Find the radius of curvature for the curve $x^4 + y^4 = 2$ at $(1, 1)$.
08. Find the radius of curvature for the curve $x^2 y = a(x^2 + y^2)$ at the point $(-2a, 2a)$.

09. Find the radius of curvature of the curve
 $x = a \left[\cos t + \log \tan \frac{t}{2} \right] \quad y = a \sin t$

Module - 02

01. Expand $\log(1 + \cos x)$ by the Maclaurin's series up to the term containing x^4 .
02. Expand $\log(1 + \sin x)$ by the Maclaurin's series up to the term containing x^4 .
03. Expand $\log(1 + e^x)$ using Maclaurin's series up to and including 3rd degree term.
04. Evaluate the following intermediate forms:

(i) $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^2} \quad (ii) \lim_{x \rightarrow 0} \left(\frac{a^x + b^x}{2} \right)^{1/x}$

(iii) $\lim_{x \rightarrow a} \left(2 - \frac{x}{a} \right)^{\tan\left(\frac{\pi x}{2a}\right)} \quad (iv) \lim_{x \rightarrow 0} (\cos x)^{1/x^2}$

05. If $u = x^2y + y^2z + z^2x$, then show that
 $u_x + u_y + u_z = (x + y + z)^2$

06. If $u = (x^2 + y^2 + z^2)^{-1/2}$, then prove that
 $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$.

07. If $u = f(x-y, y-z, z-x)$ show that
 $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

08. If $u = f(2x-3y, 3y-4z, 4z-2x)$ then find
 $\frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z}$.

09. Examine the function $f(x, y) = 2 + 2x + 2y - x^2 - y^2$ for its extreme values.

10. Show that the function $xy(a-x-y)$ is maximum at $(\frac{a}{3}, \frac{a}{3})$. Hence find maximum value if $a > 0$.

11. Show that the function $f(x, y) = x^3 + y^3 - 63x - 63y + 12xy$ is maximum at $(-7, -7)$

12. If $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$ and $z = r \cos \theta$ find $\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)}$.

13. If $x + y + z = u$, $y + z = uv$, $z = uvw$, then evaluate $\frac{\partial(x, y, z)}{\partial(u, v, w)}$.

14. If x, y, z are angles of a triangle, show that the maximum value of $\sin x \sin y \sin z$ is $\frac{3\sqrt{3}}{8}$.

Module - 03

01. Solve the following differential equations:

(i) $\frac{dy}{dx} + y \tan x = y^2 \sec x$.

(ii) $(x^2 + y^3 + 6x)dx + y^2 x dy = 0$

(iii) $\frac{dy}{dx} + \frac{y}{x} = y^2 x$

(iv) $(4xy + 3y^2 - x)dx + x(x + 2y)dy = 0$

02. Find the orthogonal trajectories of the curve $r^n \cos n\theta = a^n$.

03. Find the orthogonal trajectories of the curve $r = 4a \sec \theta \tan \theta$.

04. Solve $P(P+y) = x(x+y)$.
05. Solve $yp^2 + (x-y)p - x = 0$
06. Solve $xp^2 - (2x+3y)p + 6y = 0$
07. Solve $p = \sin(y-xp)$.
08. A series circuit with resistance R , inductance L and electromotive force E is governed by the differential equation $L \frac{di}{dt} = Ri = E$, where L and R are constants and initially the current i is zero. Find the current at any time t .

Module - 05

01. Find the rank of the following matrices by reducing it to echelon form:

(i)
$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$$

(ii)
$$\begin{bmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 6 & 8 \\ 4 & 8 & 12 & 16 \\ 1 & 2 & 3 & 4 \end{bmatrix}$$

(iii)
$$\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

(iv)
$$\begin{bmatrix} -2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & -1 \end{bmatrix}$$

02. Find the value of λ and μ such that the system $x+2y+3z=6$; $x+3y+5z=9$; $2x+5y+\lambda z=\mu$ have (i) no solution (ii) a unique solution (iii) an infinite number of solutions?

03. Show that the following system is consistent and solve:

$$x + y + z = 4 ; 2x + y - z = 1 ; x - 2y + 2z = 2 .$$

04. Test for consistency and solve:

$$5x + 3y + 7z = 4 ; 3x + 26y + 2z = 9 ; 7x + 2y + 10z = 5$$

05. Solve the following system of equations by Gauss Elimination method:

(i) $x - 2y + 3z = 2$, $3x - y + 4z = 4$, $2x + y - 2z = 5$

(ii) $2x - 3y + 4z = 7$, $5x - 2y + 2z = 7$, $6x - 3y + 10z = 23$

(iii) $2x - y + 3z = 1$, $-3x + 4y - 5z = 0$, $x + 3y + 6z = 0$

06. Solve the following system of equations by Gauss Jordan method:

(i) $2x_1 + x_2 + 3x_3 = 1$, $4x_1 + 4x_2 + 7x_3 = 1$,

$$2x_1 + 5x_2 + 9x_3 = 3 .$$

(ii) $2x - 3y + z = 1$, $x + 4y + 5z = 25$, $3x - 4y + z = 2$.

07. Solve the following system of equations by Gauss Seidel method performing 3 iterations.

(i) $3x + 8y + 29z = 71$; $83x + 11y - 4z = 95$;

$$7x + 52y + 13z = 104 .$$

(ii) $x + y + 54z = 110$, $27x + 6y - z = 85$, $6x + 15y + 2z = 72$

08. Find the largest Eigen value and the corresponding Eigen vector of the following matrices using power method:

(i) $A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ Take $(1, 1, 0)$ as initial vector. Carry out six iterations.

- (ii) $A = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$ Take $(1, 0, 0)$ as initial vector. Carry out seven iterations.

Module - 04 [Physics cycle]

01. Find the last digit of the following:

(i) 7^{2013}

(ii) 13^{37} .

02. Find the least positive values of x such that $89 \equiv (x+3) \pmod{4}$.

03. Solve $4x+3y \equiv 2 \pmod{7}$

04. Find the remainder when the number 2^{1000} is divided by 13.

05. Find the remainder when $(349 \times 74 \times 36)$ is divided by 3.

Module - 04 [Chemistry cycle]

01. Evaluate $\int_{-c}^c \int_{-b}^b \int_{-a}^a (x^2 + y^2 + z^2) dx dy dz$

02. Evaluate $\int_0^1 \int_0^{\sqrt{1-y^2}} (x^2 + y^2) dx dy$ by changing to polar co-ordinates.

03. Prove that $\int_0^{\pi/2} \sqrt{\sin \theta} d\theta \times \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} = \pi$

04. Change the order of integration and evaluate $\int_0^1 \int_{\sqrt{y}}^y dx dy$.